

DATA ANALYSIS PROJECT REPORT

How Weather, Time, and Holidays Affect Public Bike Rental Demand in Seoul

Item	Details
Project Title	How Weather, Time, and Holidays Affect Public Bike Rental Demand in Seoul
Student Name	Nurul Azam Bhuiyan
Student ID	23-50020-1
Course / Section	Programming in Python, Section C
Instructor	MD. TANZEEM RAHAT
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Prepared as a mini data investigation using Pandas, NumPy, and Matplotlib.

1. Executive Summary

This project analyzes the Seoul Bike Sharing Demand dataset to understand how hourly bike rentals change with time, weather, holidays, and service availability. The dataset contains 8,760 hourly records and 14 columns from a real public bike sharing context. Pandas was used for loading, cleaning, feature engineering, filtering, grouping, and aggregation. NumPy was used for percentile based demand levels, z score outlier detection, mean, standard deviation, and IQR calculations. Matplotlib was used to create visual evidence for the main analytical questions. The strongest results show that demand is highest in Summer with an average of 1034.07 rentals, highest by month in June with 1245.68 rentals on average, and highest by hour at 18:00 with 1554.02 rentals on average. Weather also matters: dry conditions have the highest average demand, and temperature is the strongest original weather variable associated with rentals.

2. Introduction and Project Objectives

Bike sharing systems are useful for short distance city travel, but operators must prepare enough bikes during high demand periods and avoid wasting resources during low demand periods. This project investigates hourly bike rental records from Seoul to identify demand patterns connected with time, weather, holiday status, and unusual high or low demand hours.

Item	Content
Dataset name	Seoul Bike Sharing Demand
Dataset Source	UCI Machine Learning Repository
Source Link	https://archive.ics.uci.edu/dataset/560/seoul+bike+sharing+demand
Main project goal	Analyze how time, weather, holidays, and service availability affect hourly bike rental demand in Seoul.
Research Question 1	Which time related factors, such as hour, month, season, and weekend status, have the highest bike rental demand?
Research Question 2	How do weather factors such as temperature, humidity, rainfall, snowfall, and solar radiation relate to bike rental demand?
Research Question 3	Are there unusual high demand or low demand hours, and what do those patterns tell us?

3. Dataset Description

Each row in the dataset represents one hour of bike rental activity in Seoul. The main variable is rentedBikeCount. Supporting variables include date, hour, temperature, humidity, wind speed, visibility, dew point temperature, solar radiation, rainfall, snowfall, season, holiday status, and whether the bike sharing service was functioning. The dataset is useful because it contains numerical, categorical, and date-time variables.

Property	Value
Number of rows	8,760 original rows; 8,465 service-open rows used for main demand analysis
Number of columns	14 original columns; 25 columns after feature engineering
Data types present	Numerical, categorical, and date-time variables
Key variable	rentedBikeCount
First-glance issues	Special symbols in column names, service-closed zero rentals, and possible high demand outliers

4. Data Cleaning and Preparation

The cleaning approach was designed to keep the dataset understandable while avoiding misleading values. The main cleaning actions are summarized below.

Issue Found	Action Taken	Reason / Justification	Code Reference
Column names had spaces and special symbols	Renamed columns to simple camelCase names	Simpler names reduce coding mistakes and make the notebook easier to explain.	Section 4
Possible duplicate hourly rows	Checked and removed exact duplicates	Duplicate rows could count the same hour more than once.	Section 5
Date and numeric values needed correct types	Converted date with <code>pd.to_datetime</code> and numerical columns with <code>pd.to_numeric</code>	Correct types are required for grouping, statistics, correlation, and plotting.	Section 6
Text labels could contain inconsistent formatting	Standardized season, holiday, and <code>functioningDay</code> labels	Consistent labels prevent incorrect subgroup results.	Section 7
Invalid or missing key values could weaken analysis	Dropped rows without date or rental count and checked valid hour range	Rows without key values cannot answer the research questions.	Section 8
Service-closed rows show zero rentals because bikes were unavailable	Created <code>serviceStatus</code> and used only <code>functioningDay = Yes</code> for main demand analysis	Closed-service zeros are operational unavailability, not natural customer demand.	Section 9

5. Feature Engineering

Feature engineering created new columns that directly support the research questions. The most important derived features are listed below.

New Feature	How It Was Created	Why It Is Useful
<code>monthNumber / monthName</code>	Extracted from the date column	Supports month by month demand analysis.
<code>weekendStatus</code>	Classified Saturday and Sunday as Weekend, others as Weekday	Supports weekday versus weekend comparison.
<code>timePeriod</code>	Grouped hours into Night, Morning, Afternoon, and Evening	Makes hourly behavior easier to explain.
<code>weatherCondition</code>	Used rainfall and snowfall to classify Dry, Rainy, Snowy, or Rain And Snow	Supports weather based subgroup comparison.
<code>temperatureGroup</code>	Grouped temperature into Cold, Comfortable, and Hot	Shows demand differences across understandable temperature levels.
<code>demandLevel</code>	Used NumPy percentiles to classify Low, Medium, and High Demand	Supports demand segmentation and unusual pattern analysis.
<code>weatherComfortScore</code>	100 minus temperature, humidity, rain, and snow penalties	Creates a simple custom indicator for overall riding comfort.

6. Analysis Methodology

- Pandas was used to load the dataset, inspect data quality, rename columns, clean records, create derived columns, group data, and generate summary tables.
- NumPy was used for percentile based demand classification, mean, standard deviation, z score outlier detection, and IQR based outlier limits.
- Matplotlib was used to create visualizations for time patterns, weather relationships, and demand levels.
- The main analysis used only service-open rows because service-closed zero values represent system unavailability rather than natural customer demand.

7. Analysis and Visualizations

7.1 Research Question 1: Time related demand patterns

The analysis compared demand by season, month, hour, weekend status, holiday status, and time period. The strongest time related patterns were that Summer had the highest average seasonal demand (1034.07), June had the highest monthly average (1245.68), and hour 18 had the highest average hourly demand (1554.02). Weekdays also had higher average demand than weekends (748.11 versus 682.38).

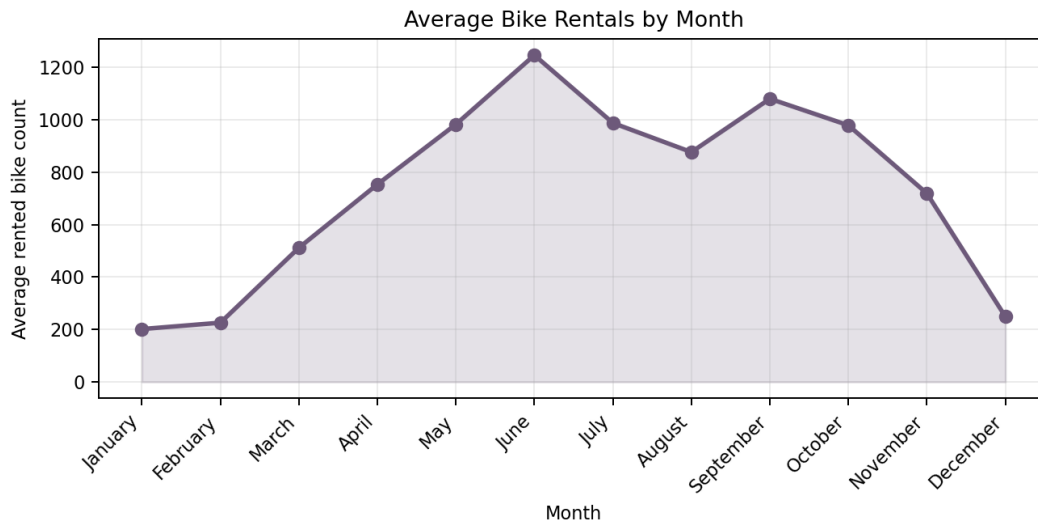


Figure 1. Average bike rentals by month. Demand rises strongly from spring, peaks in June, and falls in winter.

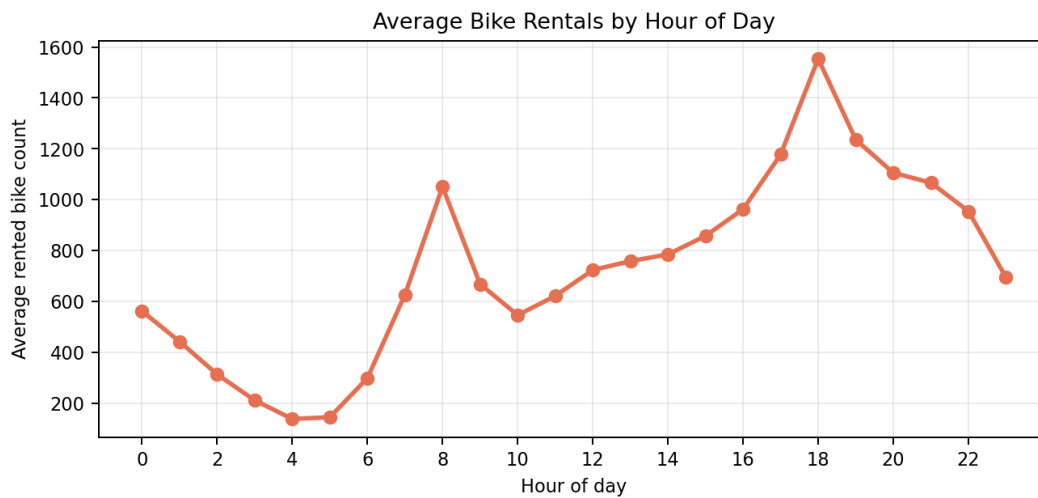


Figure 2. Average bike rentals by hour of day. The highest average demand appears at 18:00, suggesting a strong evening peak.

7.2 Research Question 2: Weather and rental demand

Weather based grouping showed that dry hours had the highest average rental demand (797.28). Rainy, snowy, and mixed rain-snow periods had much lower average demand, which supports the idea that poor weather discourages bike usage. The correlation table showed that temperature was the strongest original weather variable associated with rental count (correlation 0.5627), while humidity had a negative relationship with demand. The custom weatherComfortScore had a slightly higher correlation (0.5763), but it was treated as an approximate custom indicator rather than a scientific formula.

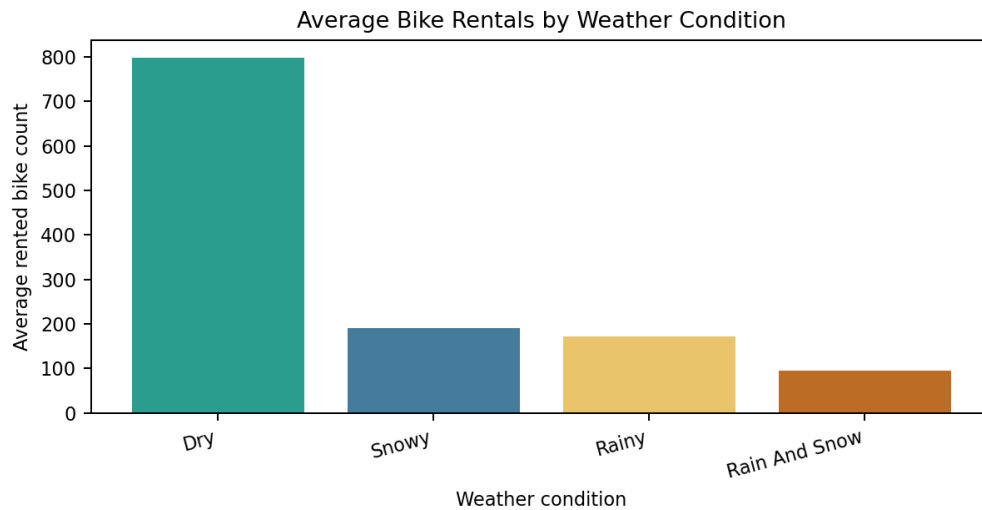


Figure 3. Average rentals by weather condition. Dry weather has the highest average demand.

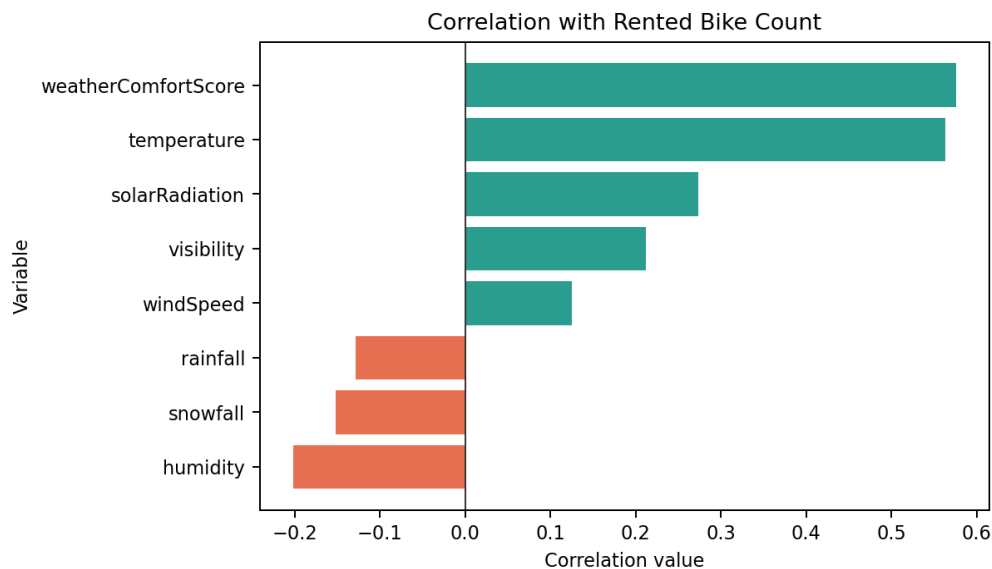


Figure 4. Correlation between weather variables and rented bike count. Temperature is the strongest original weather variable.

7.3 Research Question 3: Unusual demand patterns and outliers

Unusual demand analysis was performed using top and bottom records, NumPy z scores, and IQR limits. The highest single demand record occurred on 2018 06 19 at hour 18 with 3556 rentals. The lowest service open record occurred on 2018 10 06 at hour 10 with only 2 rentals. NumPy z score analysis identified 63 very high demand records, while IQR analysis identified 152 high demand outliers.

The season and hour heatmap adds more context by showing that demand is not equally distributed across the day or across seasons. Stronger demand appears mainly during warmer seasons and peak travel hours, especially evening periods. This supports the idea that unusual high demand is connected with both time and seasonal conditions.

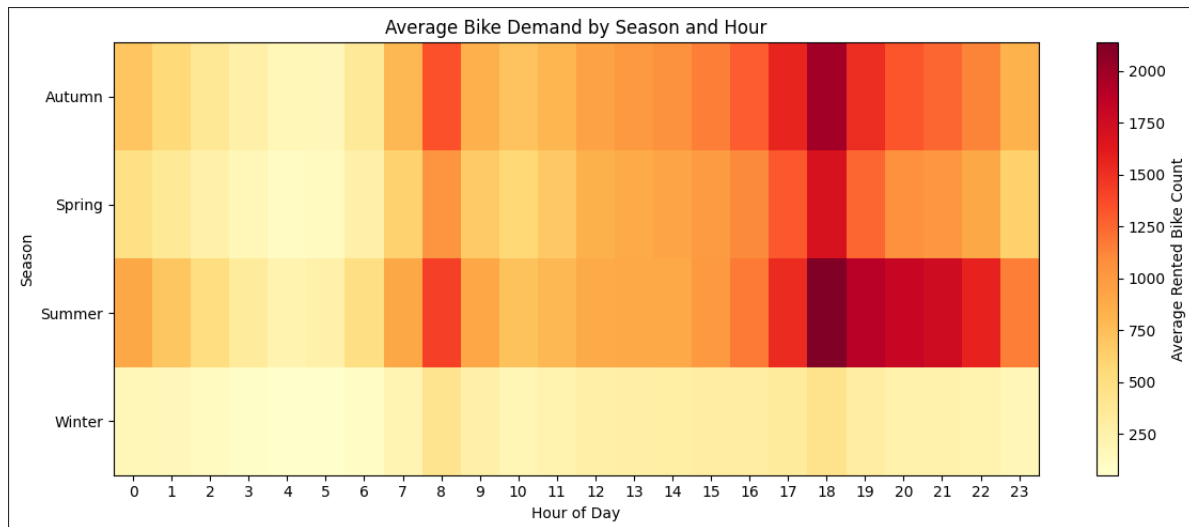


Figure 5. Season Hour Demand Heatmap. Stronger color means higher average bike rental demand

8. NumPy-Based Custom Computation

Component	Description
Computation used	Percentiles for demandLevel; mean and standard deviation for z score; percentile based quartiles for IQR.
Purpose	To classify demand levels and identify unusual high demand hours using numerical rules.
Formula or logic	$z \text{ score} = (\text{rentedBikeCount} - \text{mean}) / \text{standard deviation}$; IQR high limit = $Q3 + 1.5 \times \text{IQR}$.
Result / takeaway	z score found 63 very high demand records. IQR found 152 high demand outliers.

9. Key Findings

- Summer has the highest average seasonal demand at 1034.07 rentals, while winter has much lower demand, showing a strong seasonal effect.
- June has the highest monthly average demand at 1245.68 rentals, suggesting that early summer conditions are especially suitable for bike usage.
- The busiest average hour is 18:00 with 1554.02 rentals, showing a clear evening peak that may relate to commuting or evening travel.
- Dry weather has the highest average demand at 797.28 rentals, while rainy and snowy conditions show much lower averages.
- Temperature is the strongest original weather variable associated with demand, with a correlation of 0.5627.
- Weekdays have higher average demand (748.11) than weekends (682.38), suggesting regular work or school travel influences bike use.
- Non-holiday periods have higher average demand (739.28) than holidays (529.15).
- The highest single demand record was 3556 rentals, and z score analysis found 63 very high demand outlier hours.

10. Limitations

- The dataset covers Seoul only, so the results may not directly apply to other cities with different weather, transport systems, or travel behavior.
- The dataset covers about one year, so long term changes across many years cannot be studied.
- Important variables such as station location, user type, traffic conditions, public transport disruptions, and special city events are not included.
- Correlation does not prove causation. Weather variables may be related to demand, but they are not the only reason demand changes.
- Service-closed hours were excluded from the main demand analysis because those zero values came from system unavailability, not natural customer demand.

- The weather comfort score is a simple custom feature, not a scientific formula.
- The temperature scatter plot in the notebook uses a random sample for readability, so it is a visual summary rather than a display of every record.

11. Conclusion

This project shows that Seoul bike rental demand is strongly connected with time, weather, and service availability. Demand is higher during warmer months, reaches its highest monthly average in June, and peaks most clearly at 18:00. Weather also plays an important role because dry conditions and warmer temperature are associated with higher bike rentals, while rain, snow, and high humidity are connected with lower usage.

The analysis also shows that weekdays and non holiday periods have higher average demand than weekends and holidays. NumPy based z score analysis and IQR analysis helped identify unusual high demand hours, showing that extreme demand periods are rare but important for operational planning. Overall, the findings suggest that bike sharing operators can prepare bike availability more effectively by paying attention to month, hour, weather condition, and unusually high demand periods. Future work could improve this study by adding station level location, user type, special event data, public transport information, and multiple years of records.

12. References / Dataset Source

- UCI Machine Learning Repository. Seoul Bike Sharing Demand dataset.
<https://archive.ics.uci.edu/dataset/560/seoul+bike+sharing+demand>
- Project notebook: Seoul Bike Sharing Demand Data Analysis Project with Pandas, NumPy, and Matplotlib.

13. Appendix: Summary Table

Metric	Value
Original dataset shape	8,760 rows and 14 columns
Service-open records used in main analysis	8,465 rows
Highest average season	Summer (1034.07)
Highest average month	June (1245.68)
Highest average hour	18:00 (1554.02)
Strongest original weather variable	temperature (0.5627)
Very high demand z score outliers	63
High demand IQR outliers	152

Final Submission Checklist

- ☒ I included my GitHub repository link or Google Colab notebook link in the Google Sheet provided by the instructor.
- ☒ I attached the project report as a separate file.
- ☒ My report includes dataset source, project objective, cleaning steps, feature engineering, analysis, visualizations, findings, limitations, and conclusion.
- ☒ My code uses Pandas, NumPy, and Matplotlib meaningfully.
- ☒ All bracketed placeholder text and helper notes were replaced or deleted before submission.